



Important notes / Exam rules

Answer all the questions by choosing the correct answer/writing the correct answer. There is only one correct answer for each question. Each correct answer would fetch **0.5 mark** and no partial credit.

Q 1 Which of the following sets of vectors does not span $\mathbb{R}^3$?

☐ $\{(2, -1, 0), (0, 3, 1), (1, 1, 1)\}$

☐ $\{(1, 0, 0), (0, 1, 0), (0, 1, 1)\}$

☒ $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$

☐ $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$

Q 2 Which of the following statements is true?

☐ Rank of a matrix equals the sum of the row rank and column rank of the matrix.

☐ Row rank of a matrix equals the nullity of the matrix.

☒ Row rank of a matrix equals the column rank of the matrix.

☐ Column rank of a matrix equals the nullity of the matrix.

Q 3 Which of the following is a subspace of $\mathbb{R}^2$?

☒ Set of all $(x, y) \in \mathbb{R}^2$ such that $x^2 + y^2 = 0$.

☐ Set of all $(x, y) \in \mathbb{R}^2$ such that $x^2 - y^2 = 1$.

☐ Set of all $(x, y) \in \mathbb{R}^2$ such that $x^2 + y^2 \leq 1$.

☐ Set of all $(x, y) \in \mathbb{R}^2$ such that $x^2 + y^2 = 1$.

Q 4 Let $V = \{A \in M_{5 \times 7}(\mathbb{R}) : a_{ij} = 0, \text{ if } i + j = 10 \text{ where } 1 \leq i \leq 5 \text{ and } 1 \leq j \leq 7\}$ be a vector space. Then the dimension of V is **32**.

Q 5 Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(1, 0, 0) = (1, 2)$, $T(0, 1, 0) = (-1, 1)$, and $T(0, 0, 2) = (4, 0)$. Then $T(x, y, z) = \underline{\underline{(x-y+2z, 2x+y)}}$.

Q 6 Determine the value of k such that the system of equations $x + 2y + kz = 6$, $4x + 8y + 8z = 12$ has no solution.

☒ 2.

☐ 1.

☐ 3.

☐ 0.

Q 7 Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + y, y + z, z + x)$. What is the matrix representation of T with respect to the same ordered basis $\{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ for both domain and codomain?

☐ $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

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Q 8 Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix}$. Pick the wrong statement.

☒ The row vectors of A span $\mathbb{R}^3$.

☐ The rank of A is 2.

☐ The system of equations $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

☐ The column vectors of A span $\mathbb{R}^2$.

Q 9 What is the rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix}$?

☐ 3

☐ 0

☐ 2

☒ 1

Q 10 What is the value of a such that the rank of the matrix $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 3 & -1 & 0 \\ 1 & 3 & a & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$ is 3?

☐ 2☐ 3☐ 0☒ -1

Q 11 Which of the following statements is false?

☐ An inconsistent system has no solution.☒ A consistent system may have exactly two solutions.☐ A consistent system may have a unique solution.☐ A consistent system may have infinite solutions.

Q 12 Consider the system: $x + 3y - 2z = 0$, $-4x - 12y + 8z = 5$, $5x + 15y - 10z = 7$. The rank of the coefficient matrix is

☐ 3.☐ 2.☒ 1.☐ 0.

Q 13 The set $S = \{(a, 1, 0), (1, a, 1), (0, 1, a)\}$ forms a basis of $\mathbb{R}^3$ if

☐ $a = 0$ ☒ $a \neq 0$ and $a^2 \neq 2$ ☐ $a \neq 0$ and $a^2 = 2$ ☐ $a \neq 0$

Q 14 Let $S = \{(x, y, z) \in \mathbb{R}^3 : 5x - 3y = 0, 5x - y + 3z = 0, 2y + 3z = 1\}$. Then

☒ S is not a subspace of $\mathbb{R}^3$ ☐ S is a subspace of $\mathbb{R}^3$ with $\dim(S) = 2$ ☐ S is a subspace of $\mathbb{R}^2$ with $\dim(S) = 2$ ☐ S is a subspace of $\mathbb{R}^3$ with $\dim(S) = 3$

Q 15 Let S and T be two subspaces of a vector space V over $\mathbb{R}$. Then which of the following is always true?

☐ $S \cup T$ is a subspace of V .☒ $S \cap T$ is a subspace of V .☐ $S \cap T$ is a subspace of V only if $S \cap T = \emptyset$.☐ None of the other options.

Q 16 Let c and n be the dimension of the column space and null space of the matrix $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}$. Which of the following is correct?

☐ $c = 1$ and $n = 2$.☐ $c = 1$ and $n = 1$.☒ $c = 2$ and $n = 2$.☐ $c = 2$ and $n = 1$.

Q 17 Let $T : \mathbb{M}_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{M}_{2 \times 2}(\mathbb{R})$ be a linear transformation defined as $T(A) = A + A^T$. Then

☐ $\text{Rank}(T) = 2$ and $\text{Nullity}(T) = 2$ ☐ $\text{Rank}(T) = 2$ and $\text{Nullity}(T) = 1$ ☒ $\text{Rank}(T) = 3$ and $\text{Nullity}(T) = 1$ ☐ $\text{Rank}(T) = 1$ and $\text{Nullity}(T) = 3$

Q 18 Which of the following sets is not a vector space over $\mathbb{R}$ with the standard addition and scalar multiplication?

☐ Set of all continuous functions on the interval $[0, 5]$.☐ Set of all 3×3 real symmetric matrices.☒ Set of all $(x, y, z) \in \mathbb{R}^3$ such that $xy = 0$.☐ Set of all polynomials of degree less or equal to 5.

Q 19 Consider the system: $x + 3y - 2z = 0$, $-4x - 12y + 8z = 5$, $5x + 15y - 10z = 7$. Pick the correct statement.

☐ The system of equations has infinitely many solutions.☐ The system of equations has a unique solution.☒ The system of equations has no solution.☐ None of the other options.

Q 20 Let S be the vector space of all real polynomials $p(t)$ such that the second derivative $p''(t) = 0$ for all $t \in \mathbb{R}$. Then the dimension of S is

☐ 1☐ 3☒ 2☐ infinite

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Section:.....

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☐ infinite

☐ 1

☐ 3

☒ 2

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☐ S is a subspace of $\mathbb{R}^3$ with $\dim(S) = 3$ ☐ S is a subspace of $\mathbb{R}^3$ with $\dim(S) = 2$
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☐ The rank of A is 2. ☒ The row vectors of A span $\mathbb{R}^3$.
☐ The column vectors of A span $\mathbb{R}^2$. ☐ The system of equations $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

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☐ $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ ☐ $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ ☒ $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$ ☐ $\begin{bmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

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Q 17 Let S and T be two subspaces of a vector space V over $\mathbb{R}$. Then which of the following is always true?

☐ None of the other options.☒ $S \cap T$ is a subspace of V .☐ $S \cap T$ is a subspace of V only if $S \cap T = \emptyset$.☐ $S \cup T$ is a subspace of V .

Q 18 The set $S = \{(a, 1, 0), (1, a, 1), (0, 1, a)\}$ forms a basis of $\mathbb{R}^3$ if

☒ $a \neq 0$ and $a^2 \neq 2$ ☐ $a \neq 0$ and $a^2 = 2$ ☐ $a = 0$ ☐ $a \neq 0$

Q 19 Let S be the vector space of all real polynomials $p(t)$ such that the second derivative $p''(t) = 0$ for all $t \in \mathbb{R}$. Then the dimension of S is

☐ infinite☒ 2☐ 1☐ 3

Q 20 Let c and n be the dimension of the column space and null space of the matrix $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}$. Which of the following is correct?

☒ $c = 2$ and $n = 2$.☐ $c = 1$ and $n = 1$.☐ $c = 2$ and $n = 1$.☐ $c = 1$ and $n = 2$.